

Necessary and Sufficient Conditions for Local Extrema

1 Introduction and Notation

Let $U \subset \mathbb{R}^n$ be an open set. We consider scalar-valued functions $f: U \rightarrow \mathbb{R}$ of class C^2 . Vectors $x \in \mathbb{R}^n$ are treated as column vectors $x = [x_1, \dots, x_n]^\top$. When partitioning x into subvectors $y \in \mathbb{R}^{n-m}$ and $z \in \mathbb{R}^m$, we write $x = (y, z)$ as a shorthand for the block column vector $\begin{bmatrix} y \\ z \end{bmatrix}$.

The first derivative (or total differential) of f at $x \in U$ is represented by the $1 \times n$ Jacobian row vector:

$$Df(x) = \left[\frac{\partial f}{\partial x_1}(x) \quad \dots \quad \frac{\partial f}{\partial x_n}(x) \right].$$

The gradient $\nabla f(x)$ is the column vector $Df(x)^\top \in \mathbb{R}^n$. For a direction vector $v \in \mathbb{R}^n$, the directional derivative $D_v f(x)$ is given by the matrix-vector product $Df(x)v$.

The second derivative of f at $x \in U$ is represented by the $n \times n$ symmetric Hessian matrix:

$$D^2 f(x) = D(\nabla f)(x) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right]_{1 \leq i, j \leq n}.$$

The second directional derivative in the direction of $v \in \mathbb{R}^n$ is given by the quadratic form $v^\top D^2 f(x)v$.

We state three classical results from multivariable calculus that serve as foundational tools for our analysis.

Lemma 1.1 (1D Taylor's Theorem with Lagrange Remainder). *If $\phi: [0, 1] \rightarrow \mathbb{R}$ is continuous on $[0, 1]$ and twice differentiable on $(0, 1)$, there exists $c \in (0, 1)$ such that*

$$\phi(1) = \phi(0) + \phi'(0) + \frac{1}{2}\phi''(c).$$

Theorem 1.2 (Multivariable Chain Rule). *Let $A \subset \mathbb{R}^k$ and $B \subset \mathbb{R}^n$ be open sets. Let $P: A \rightarrow B$ be differentiable at $y \in A$, and let $L: B \rightarrow \mathbb{R}$ be differentiable at $P(y)$. Then the composite function $F = L \circ P: A \rightarrow \mathbb{R}$ is differentiable at y , and its derivative row vector is given by the matrix product:*

$$DF(y) = DL(P(y))DP(y).$$

Theorem 1.3 (Implicit Function Theorem). *Let $A \subset \mathbb{R}^{n-m}$ and $B \subset \mathbb{R}^m$ be open sets. Let $g: A \times B \rightarrow \mathbb{R}^m$ be of class C^k ($k \geq 1$). Suppose that at a point $(y^*, z^*) \in A \times B$, we have $g(y^*, z^*) = 0$ and the $m \times m$ partial Jacobian matrix $D_z g(y^*, z^*)$ is invertible. Then there exist open neighborhoods $W \subset A$ of y^* and $V \subset B$ of z^* , and a C^k function $\phi: W \rightarrow V$ such that $\phi(y^*) = z^*$ and, for all $(y, z) \in W \times V$,*

$$g(y, z) = 0 \iff z = \phi(y).$$

2 Taylor's Theorem in $\mathbb{R}^n$

To establish precise local optimality conditions, we rely on the second-order multivariable Taylor expansion with a Peano remainder.

Theorem 2.1 (Second-Order Taylor Expansion). *Let $U \subset \mathbb{R}^n$ be open and $f \in C^2(U, \mathbb{R})$. Let $x \in U$ and suppose $B_r(x) \subset U$. For any perturbation vector $h \in \mathbb{R}^n$ with $\|h\| < r$,*

$$f(x+h) = f(x) + Df(x)h + \frac{1}{2}h^\top D^2f(x)h + r(h),$$

where the remainder term satisfies $\lim_{h \rightarrow 0} \frac{r(h)}{\|h\|^2} = 0$.

Proof. Fix $x \in U$ and $h \in \mathbb{R}^n$ with $\|h\| < r$. Define the line segment parameterization $\gamma: [0, 1] \rightarrow U$ by $\gamma(t) = x + th$, and set $\phi(t) = f(\gamma(t))$. By the Chain Rule (Theorem 1.2), $\phi'(t) = Df(x+th)h$. Applying the Chain Rule to the entry functions of Df , we obtain $\phi''(t) = h^\top D^2f(x+th)h$.

By Lemma 1.1, there exists $c \in (0, 1)$ such that $\phi(1) = \phi(0) + \phi'(0) + \frac{1}{2}\phi''(c)$. Substituting our expressions for $\phi(1)$, $\phi(0)$, $\phi'(0)$, and $\phi''(c)$ yields:

$$f(x+h) = f(x) + Df(x)h + \frac{1}{2}h^\top D^2f(x+ch)h.$$

We define $r(h)$ by isolating the difference between the Hessian evaluated at $x+ch$ and at x :

$$r(h) = \frac{1}{2}h^\top [D^2f(x+ch) - D^2f(x)] h.$$

Because $f \in C^2$, the Hessian map $x \mapsto D^2f(x)$ is continuous. As $h \rightarrow 0$, $x+ch \rightarrow x$, so $D^2f(x+ch) \rightarrow D^2f(x)$. Applying standard matrix operator norm bounds, we obtain:

$$|r(h)| \leq \frac{1}{2}\|h\|^2 \|D^2f(x+ch) - D^2f(x)\|_{op}.$$

Dividing by $\|h\|^2$ and taking $h \rightarrow 0$ yields $\lim_{h \rightarrow 0} \frac{r(h)}{\|h\|^2} = 0$. □

3 Unconstrained Local Extrema

Definition 3.1. A point $x^* \in U$ is a **local minimum** of f if there exists $\epsilon > 0$ such that $f(x^* + h) \geq f(x^*)$ for all $h \in \mathbb{R}^n$ with $\|h\| < \epsilon$. It is a **strict local minimum** if $f(x^* + h) > f(x^*)$ for all $h \in \mathbb{R}^n$ satisfying $0 < \|h\| < \epsilon$.

Theorem 3.2 (First-Order Necessary Condition). *If x^* is a local minimum of f , then $Df(x^*) = 0$.*

Proof. Let $v \in \mathbb{R}^n$ be an arbitrary vector. For $t \in \mathbb{R} \setminus \{0\}$ sufficiently close to 0, $x^* + tv \in U$ and $f(x^* + tv) - f(x^*) \geq 0$. Evaluating directional derivatives via unilateral limits:

$$D_v f(x^*) = \lim_{t \rightarrow 0^+} \frac{f(x^* + tv) - f(x^*)}{t} \geq 0,$$

and

$$D_v f(x^*) = \lim_{t \rightarrow 0^-} \frac{f(x^* + tv) - f(x^*)}{t} \leq 0.$$

Thus, $D_v f(x^*) = Df(x^*)v = 0$ for all $v \in \mathbb{R}^n$. Setting $v = e_i$ (the standard basis vectors) proves $Df(x^*) = 0$. □

Theorem 3.3 (Second-Order Necessary Condition). *If x^* is a local minimum of f , then the Hessian matrix $D^2f(x^*)$ is positive semi-definite (i.e., $v^\top D^2f(x^*)v \geq 0$ for all $v \in \mathbb{R}^n$).*

Proof. By Theorem 3.2, $Df(x^*) = 0$. Let $v \in \mathbb{R}^n$. For $t \neq 0$ near 0, Theorem 2.1 gives:

$$f(x^* + tv) - f(x^*) = \frac{t^2}{2} v^\top D^2 f(x^*) v + r(tv) \geq 0.$$

Dividing by t^2 yields $\frac{1}{2} v^\top D^2 f(x^*) v + \frac{r(tv)}{t^2} \geq 0$. Taking $t \rightarrow 0$ (noting $\frac{r(tv)}{t^2} = \|v\|^2 \frac{r(tv)}{\|tv\|^2} \rightarrow 0$) proves $v^\top D^2 f(x^*) v \geq 0$. $\square$

Theorem 3.4 (Second-Order Sufficient Condition). *If $Df(x^*) = 0$ and $D^2 f(x^*)$ is positive definite (i.e., $v^\top D^2 f(x^*) v > 0$ for all $v \neq 0$), then x^* is a strict local minimum.*

Proof. The quadratic form $v \mapsto v^\top D^2 f(x^*) v$ is continuous. Since the unit sphere $S^{n-1} \subset \mathbb{R}^n$ is compact, the map attains a positive minimum $\lambda > 0$ on S^{n-1} . For any non-zero $v \in \mathbb{R}^n$, setting $u = v / \|v\| \in S^{n-1}$ gives $v^\top D^2 f(x^*) v = \|v\|^2 (u^\top D^2 f(x^*) u) \geq \lambda \|v\|^2$.

By Theorem 2.1 (using $Df(x^*) = 0$):

$$f(x^* + v) - f(x^*) = \frac{1}{2} v^\top D^2 f(x^*) v + r(v) \geq \|v\|^2 \left(\frac{\lambda}{2} + \frac{r(v)}{\|v\|^2} \right).$$

Since $\lim_{v \rightarrow 0} \frac{r(v)}{\|v\|^2} = 0$, there exists $\epsilon > 0$ such that for all $0 < \|v\| < \epsilon$, $\left| \frac{r(v)}{\|v\|^2} \right| < \frac{\lambda}{4}$. Consequently, $f(x^* + v) - f(x^*) \geq \frac{\lambda}{4} \|v\|^2 > 0$, confirming x^* is a strict local minimum. $\square$

4 Constrained Local Extrema: Setup and First-Order Conditions

Let $g: U \rightarrow \mathbb{R}^m$ be a function of class C^2 with $m < n$. We define the constraint set $S = \{x \in U \mid g(x) = 0\}$ and consider local extrema of f restricted to S .

4.1 Regularity and Local Parameterization

A point $x^* \in S$ is a **regular point** if the $m \times n$ Jacobian matrix $Dg(x^*)$ has full rank m . Permuting coordinates if necessary, we write $x = (y, z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$ such that the $m \times m$ partial Jacobian matrix $D_z g(x^*)$ is invertible.

By the Implicit Function Theorem (Theorem 1.3), there exist open neighborhoods $W \subset \mathbb{R}^{n-m}$ of y^* and $V \subset \mathbb{R}^m$ of z^* , and a C^2 map $\phi: W \rightarrow V$ such that $\phi(y^*) = z^*$ and:

$$g(y, \phi(y)) \equiv 0 \quad \text{for all } y \in W.$$

Differentiating this identity via the Chain Rule gives:

$$D_y g(y, \phi(y)) + D_z g(y, \phi(y)) D\phi(y) = 0.$$

Evaluating at $y = y^*$ (where $(y^*, \phi(y^*)) = x^*$), we solve for the derivative of the parameterization:

$$D\phi(y^*) = -[D_z g(x^*)]^{-1} D_y g(x^*). \quad (1)$$

S

4.2 The Tangent Space

The tangent space to S at x^* is defined linearly as $T_{x^*}S = \ker Dg(x^*)$. A vector $v = (v_y, v_z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$ belongs to $T_{x^*}S$ if and only if:

$$Dg(x^*)v = 0 \iff D_y g(x^*)v_y + D_z g(x^*)v_z = 0 \iff v_z = -[D_z g(x^*)]^{-1} D_y g(x^*)v_y.$$

Applying equation (1), this simplifies to $v_z = D\phi(y^*)v_y$. Thus, every tangent vector $v \in T_{x^*}S$ is uniquely represented by its projection $v_y \in \mathbb{R}^{n-m}$ via:

$$v = \begin{bmatrix} v_y \\ D\phi(y^*)v_y \end{bmatrix}.$$

4.3 First-Order Conditions (Lagrange Multipliers)

Theorem 4.1. *Let $x^* = (y^*, z^*) \in S$ be a regular point. If x^* is a local minimum of f on S , there exists a unique Lagrange multiplier vector $\lambda \in \mathbb{R}^m$ such that $DL(x^*) = 0$, where $L(x) = f(x) - \lambda^\top g(x)$.*

Proof. Define the reduced objective function $\tilde{f}: W \rightarrow \mathbb{R}$ by $\tilde{f}(y) = f(y, \phi(y))$. Because x^* is a constrained local minimum of f on S , y^* is an unconstrained local minimum of $\tilde{f}$ on W . Theorem 3.2 implies $D\tilde{f}(y^*) = 0$.

Applying the Chain Rule to $\tilde{f}(y)$:

$$D\tilde{f}(y^*) = D_y f(x^*) + D_z f(x^*)D\phi(y^*) = 0. \quad (2)$$

We define the vector $\lambda \in \mathbb{R}^m$ by specifying its transposed row vector:

$$\lambda^\top = D_z f(x^*)[D_z g(x^*)]^{-1}.$$

This definition guarantees that:

$$D_z L(x^*) = D_z f(x^*) - \lambda^\top D_z g(x^*) = 0. \quad (3)$$

Substituting equation (1) into (2) yields:

$$D_y f(x^*) - D_z f(x^*)[D_z g(x^*)]^{-1} D_y g(x^*) = 0 \implies D_y f(x^*) - \lambda^\top D_y g(x^*) = 0.$$

Thus $D_y L(x^*) = 0$. Combining this with (3), the block derivative satisfies $DL(x^*) = \begin{bmatrix} D_y L(x^*) & D_z L(x^*) \end{bmatrix} = 0$. $\square$

5 Constrained Local Extrema: Second-Order Conditions

To evaluate the second derivative of the reduced objective $D^2\tilde{f}(y^*)$ without calculating $D^2\phi$, we use the algebraic structure of the Lagrangian.

The Algebraic Collapse Because $g(y, \phi(y)) \equiv 0$ on W , subtracting $\lambda^\top g(y, \phi(y))$ leaves $\tilde{f}$ unchanged:

$$\tilde{f}(y) = f(y, \phi(y)) - \lambda^\top g(y, \phi(y)) = L(y, \phi(y)).$$

Define the local parameterization map $P: W \rightarrow \mathbb{R}^n$ by $P(y) = (y, \phi(y))$, so $\tilde{f}(y) = L(P(y))$.

Let $w \in \mathbb{R}^{n-m}$ be arbitrary. Consider the 1D slice function $\psi(t) = \tilde{f}(y^* + tw) = L(P(y^* + tw))$ for t near 0. Define the velocity vector $v(t) = \frac{d}{dt}P(y^* + tw) \in \mathbb{R}^n$. By the Chain Rule, $v(t) = DP(y^* + tw)w$. At $t = 0$:

$$v = v(0) = DP(y^*)w = \begin{bmatrix} w \\ D\phi(y^*)w \end{bmatrix} \in \ker Dg(x^*).$$

Differentiating $\psi(t)$ twice with respect to t via the product and chain rules gives:

$$\psi''(t) = v(t)^\top D^2L(P(y^* + tw))v(t) + DL(P(y^* + tw))v'(t).$$

Evaluating at $t = 0$, Theorem 4.1 gives $DL(P(y^*)) = DL(x^*) = 0$. The derivative term $DL(x^*)v'(0)$ vanishes completely, eliminating the second derivative $v'(0) = (0, D^2\phi(y^*)(w, w))$:

$$\psi''(0) = v^\top D^2L(x^*)v.$$

Since $\psi''(0) = D_w^2\tilde{f}(y^*) = w^\top D^2\tilde{f}(y^*)w$, we establish the foundational identity of constrained optimization:

$$w^\top D^2\tilde{f}(y^*)w = v^\top D^2L(x^*)v \quad \forall w \in \mathbb{R}^{n-m}, \quad \text{where } v = \begin{bmatrix} w \\ D\phi(y^*)w \end{bmatrix} \in \ker Dg(x^*). \quad (4)$$

Theorem 5.1 (Second-Order Constrained Necessary Condition). *Let $x^* \in S$ be a regular point. If x^* is a constrained local minimum of f on S , then $v^\top D^2L(x^*)v \geq 0$ for all $v \in \ker Dg(x^*)$.*

Proof. If x^* is a constrained local minimum, y^* is an unconstrained local minimum of $\tilde{f}$. By Theorem 3.3, $D^2\tilde{f}(y^*)$ is positive semi-definite ($w^\top D^2\tilde{f}(y^*)w \geq 0$ for all w). Identity (4) implies $v^\top D^2L(x^*)v \geq 0$ for all $v \in \ker Dg(x^*)$. $\square$

Theorem 5.2 (Second-Order Constrained Sufficient Condition). *Let $x^* \in S$ be a regular point. Suppose there exists $\lambda \in \mathbb{R}^m$ such that $DL(x^*) = 0$. If $v^\top D^2L(x^*)v > 0$ for all non-zero $v \in \ker Dg(x^*)$, then x^* is a strict constrained local minimum of f on S .*

Proof. If $v^\top D^2L(x^*)v > 0$ for all non-zero $v \in \ker Dg(x^*)$, identity (4) implies $w^\top D^2\tilde{f}(y^*)w > 0$ for all non-zero $w \in \mathbb{R}^{n-m}$. Thus, $D^2\tilde{f}(y^*)$ is positive definite. By Theorem 3.4, y^* is a strict local minimum of $\tilde{f}$ on W . Because $P(y) = (y, \phi(y))$ maps W homeomorphically onto a neighborhood of x^* in S , x^* is a strict local minimum of f on S . $\square$